

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

Programming reference generated from docs/05_ladder_logic_rK.md.

Key Rev K jog rules: any STOP press exits jog; drum OFF inhibits motion but not jog mode; the 5 s FWD entry hold is consumed by C50 and cannot command spindle motion.

Powermatic 1200 Retrofit - CLICK PLUS Ladder Logic

Transcribe into the free CLICK Programming Software. Symbols:

-] [- N.O. . -]/[- N.C. . -]UP[- leading-edge (press) . -]DN[- trailing-edge (release) .
-()- OUT . -(S)- SET . -(R)- RST . [TMR ...] timer.

> Scan order is load-bearing in Rev K. The bidirectional stop-first behavior and the jog-mode
> short-press logic depend on rungs executing in the order listed. Two rules to preserve: (a)
the DRILL
> latch-SET rungs (4, 5) run before the opposite-stop rung (7); (b) the jog/consume flag C50
> is set by the 5 s entry hold before the jog output rung can act on that held FWD input.
Transcribe in
> the numbered order and don't reorder.

RUNG 1 . RUN PERMISSIVE -> C20 (KA proved . no immediate trip . drum ON . STOP clear .
healthy)

```

X011      X013      X008      X004      C10      C11
--] [-----]/[-----+--] [-----+--] [-----]/[-----]/[----- ( )-- C20
          | X009 |
          +--] [-----+

```

C20 = X011 AND NOT X013 AND (X008 OR X009) AND X004 AND NOT C10 AND NOT C11

RUNG 2 . SPEED PRESET -> Y003 (drum LOW -> 30 Hz; jog forces LOW)

```

X008
--] [---+----- ( )-- Y003 PRESET_LOW
C60  | (jog armed -> force 30 Hz)
--] [---+

```

RUNG 3 . BLINK ASTABLE -> C99 (~1 Hz)

```

C99
--]/[-----[TMR Tb1 K500ms]
Tb1
--] [----- (S)-- C99
C99
--] [-----[TMR Tb2 K500ms]
Tb2
--] [----- (R)-- C99

```

----- DRILL - latched, bidirectional

RUNG 4 . DRILL FWD - LATCH SET on RELEASE (short tap latches; NOT C50 = not consumed by a
long-hold/opposite-stop; NOT C60 = not in jog)

```

X001      X003      C50      C60      C20      C10      C31
--] [-----]DN[-----]/[-----] [-----]/[-----]/[---(S)-- C30 DRILL_RUN
      release NOT consumed NOT jog permit noFault NOT REV

```

RUNG 5 . DRILL REV - LATCH SET on PRESS (mirror of FWD; no long-hold role, so acts on
press)

```

X001      X005      C60      C20      C10      C30
--] [-----]UP[-----]/[-----] [-----]/[-----]/[---(S)-- C31 DRILL_REV
      press NOT jog permit noFault NOT FWD

```

RUNG 6 . DRILL - LATCH RESET (stop / not DRILL / lost permit / fault -> drop both)

```

X004
--]/[---+-----+----- (R)-- C30
X001 | +----- (R)-- C31
--]/[---+ (NOT DRILL)
C20  | (NOT permit - incl. X011 safety, drum-OFF, over-temp)
--]/[---+
C10  | (fault)
--] [---+

```

RUNG 7 . OPPOSITE-AS-STOP (opposite button = stop only; runs AFTER the SET rungs)

```

X005      C30
--]UP[-----] [-----+-(R)-- C30 REV press stops a FWD run
          +-(S)-- C50 ...and consume, so its release can't
          latch REV path

X003      C31
--]UP[-----] [-----+-(R)-- C31 FWD press stops a REV run
          +-(S)-- C50 ...and consume, so its RELEASE can't
          latch FWD

```

(Why it works: RUNG 5's NOT C30 / RUNG 4's NOT C31 interlocks block the opposite latch on the
same press
because the running latch is still set when SET evaluates; RUNG 7 then drops it. Second
press latches.)

```

RUNG 8 · CLEAR FWD-CONSUMED ON RELEASE
  X003
  --]/[------(R)-- C50

RUNG 9 · DRILL BOTTOM-LIMIT FAULT   (FWD only; reverse is not a feed-down runaway)
  X001      C30      X006
  --] [-----] [-----]/[------(S)-- C10  FAULT

----- TAP - single-tap state machine
-----

RUNG 10 · TAP START (IDLE -> TAP_DOWN)  entry gate: lever OFF (X010=1), not in jog
  X002      X003      X010      C60      C20      C10      C2      C3
  --] [----]UP[----] [----]/[----] [----]/[----]/[----]/[----(S)-- C2  TAP_DOWN
                        NOT jog      permit  noFault

RUNG 11 · TAP DOWN -> BACK_OUT   (bottom limit)
  C2      X006
  --] [-----]/[-----+------(R)-- C2
                        +------(S)-- C3  BACK_OUT

RUNG 12 · TAP BACK_OUT -> IDLE   (top limit)
  C3      X007
  --] [-----]/[------(R)-- C3

RUNG 13 · TAP ABORT   (stop / not TAP / fault / lost-permit-or-safety) ← Rev H: adds NOT C20
  leg
  X004
  --]/[---+-----+------(R)-- C2
  X002 |                      +------(R)-- C3
  --]/[---+ (NOT TAP)
  C10 | (fault)
  --] [---+
  C20 | (lost permit - incl. E-stop/safety via X011, drum-OFF, STOP)
  --]/[---+

RUNG 14 · TAP FEED-LEVER RUNTIME TRIP   (lever ON during a cycle = tap-snap)
  C2      X010
  --] [---+--]/[------(S)-- C10  FAULT
  C3      | LeverON
  --] [---+

----- DB over-temp - self-clearing -----

RUNG 15 · DB OVER-TEMP SET   (X012=0 = over-temp OR wire break -> assert immediately)
  X012
  --]/[------(S)-- C11  THERMAL

RUNG 16 · DB COOL-DOWN -> SELF-CLEAR   (clears only after X012 stays normal for the cool-down
  delay)
  X012
  --] [-----[TMR Tcool K10000ms]
  Tcool
  --] [------(R)-- C11
  (Self-clearing: no manual reset, and C11 is NOT the operator-clearable C10 - so it can't
  fight RUNG 17.)

RUNG 17 · FAULT RESET   (STOP while fully stopped clears the hard-fault latch)
  X004      C2      C3      C30      C31
  --]/[-----]/[-----]/[-----]/[-----]/[------(R)-- C10  FAULT

----- JOG mode -----

RUNG 18 · FWD HOLD TIMER   (times how long FWD is held)
  X003
  --] [-----[TMR T5FWD K5000ms]

RUNG 19 · ENTER JOG   (FWD held >=5 s, safe, not already armed; consume the press so release
  won't latch a run)
  T5FWD      C50      C60      X011      X013      C10
  --] [----]/[-----]/[-----] [----]/[----]/[----+-(S)-- C60  JOG_MODE
                        +-(S)-- C50  (consumed)

RUNG 20 · JOG IDLE TIMEOUT   (60 s with no jog button -> auto-exit)
  C60      X003      X005
  --] [----]/[-----]/[-----[TMR T60 K60000ms]
  T60
  --] [------(R)-- C60

RUNG 21 · JOG FORCE-EXIT   (any latched fault, thermal trip, or safety trip)
  C10
  --] [---+------(R)-- C60
  C11 |
  --] [---+ (DB thermal trip/cooldown)
  X011 |
  --]/[---+ (final safe state lost)
  X013 |
  --] [---+ (immediate safety trip)

```

RUNG 22 · STOP EXIT (jog) (any STOP press exits jog mode)

```

C60      X004
--] [----]/[-----](R)-- C60

```

RUNG 23 · RESERVED (the former STOP-hold distinction remains deleted)

RUNG 23A · JOG RUN PERMISSIVE -> C61

```

C60      X011      X013      X008      X004      C10      C11
--] [----] [-----]/[-----+---] [-----+---] [-----]/[-----]/[-----] ( )-- C61
                        |      X009      |
                        +---] [-----+

```

----- VFD run outputs -----

RUNG 24 · VFD RUN-FWD -> Y001

```

C30      C20      Y002
--] [---+---] [-----]/[-----] ( )-- Y001 VFD_FWD
C2      | (normal: DRILL_RUN or TAP_DOWN)
--] [---+
C61      | X003      C50      Y002
--] [---+---] [-----]/[-----]/[--- (jog-FWD: NOT C50 means the 5 s entry hold can NOT jog)

```

Y001 = [(C30 OR C2) AND C20 AND NOT Y002] OR [C61 AND X003 AND NOT C50 AND NOT Y002]

RUNG 25 · VFD RUN-REV -> Y002

```

C31      C20      Y001
--] [---+---] [-----]/[-----] ( )-- Y002 VFD_REV
C3      | (normal: DRILL_REV or TAP back-out)
--] [---+
C61      | X005      Y001
--] [---+---] [-----]/[--- (jog-REV)

```

Y002 = [(C31 OR C3) AND C20 AND NOT Y001] OR [C61 AND X005 AND NOT Y001]

----- Chunk (Contactor B, theater) -----

RUNG 26 · CHUNK REQUEST (edge of entering any run / direction - TAP or DRILL)

```

C2
--]UP[---+-----](S)-- C40 CHUNK_REQ
C3      |
--]UP[---+
C30      |
--]UP[---+
C31      |
--]UP[---+

```

RUNG 27 · CHUNK TIMER + OUTPUT

```

C40
--] [-----][TMR T1 K300ms]
C40      T1
--] [-----]/[-----] ( )-- Y004 CONTACTOR_B
T1
--] [-----](R)-- C40

```

----- Indicators (4 states, exactly one active) -----

RUNG 28 · GREEN -> Y005 (solid = all-good; blink = jog armed)

```

X011      X013      C10      C11      C60
--] [-----]/[-----]/[-----]/[-----]/[-----] ( )-- Y005 GREEN
X011      | X013      C10      C11      C60      C99
--] [---+---]/[-----]/[-----]/[-----] [-----] [--- (jog: blink via C99)

```

RUNG 29 · RED -> Y006 (solid = safety/hard fault; blink = thermal trip/cooldown)

```

X013
--] [---+-----] ( )-- Y006 RED
C10      |
--] [---+ (immediate safety trip OR hard fault -> solid)
X011      C11
--]/[-----]/[---+ (KA not proved, except while thermal state explains its intentional dropout)
C11      X013      C10      C99
--] [-----]/[-----]/[-----] [-----] + (thermal blink when no solid-red cause)

```

RED_SOLID = X013 OR C10 OR (NOT X011 AND NOT C11)

RED_BLINK = C11 AND NOT X013 AND NOT C10 AND C99